Afeka College of Engineering, Tel-Aviv

Smart Thermal System for Patient Safety Monitoring

Human Detection — Algorithm Evaluation Report

MLX90640 Thermal Sensor, 22 Scenes, 80/10/10 Train/Val/Test Split

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June 2026

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1 Overview

This report presents a systematic evaluation of four human detection algorithms applied to thermal imagery from the MLX90640 32×24 sensor. Two evaluation criteria are applied to each algorithm: a lenient *frame-level presence* criterion and a strict *localisation* criterion requiring IoU > 0.5 between predicted and ground-truth bounding boxes. The goal is to characterise which algorithms are suitable for each downstream task in the safety monitoring pipeline.

2 Dataset and Experimental Setup

2.1 Dataset

Recordings were collected at the Beer Sheva Mental Health Center using three MLX90640 sensors mounted in the ward. Each sensor produces 32×24 thermal frames at 8 Hz. The evaluation uses **22 scenes** (two scenes — `2pplrun` and `3pplfight` — were excluded pending annotation review). Scene names describe the scenario: single or multiple persons, presence of a heat source, running, falling, etc.

Preprocessing is applied uniformly across all algorithms: each channel is passed through a **Tateno pipeline** (Gaussian denoising → background subtraction → L1 rectification), calibrated on even-indexed frames from the empty-room recording.

2.2 Train / Validation / Test Split

For each *(scene, channel)* group, labeled frame indices are sorted by temporal order and split sequentially: the first 80 % form the training set, the next 10 % the validation set, and the final 10 % the test set. Sequential splitting prevents temporal leakage while ensuring every scene contributes proportionally to every split.

Table 1: Split statistics (frame-channel examples).

Split	**Total**	**Person-present (Positive)**	**Empty (Negative)**
Train	4,632	4,393	239
Validation	576	559	17
Test	621	608	13

The dataset is strongly imbalanced ($\approx$ 98 % positive), reflecting real-world conditions in a monitored ward where persons are present nearly continuously.

3 Metric Definitions

3.1 Confusion Matrix

Every frame is reduced to a binary verdict: **Human Detected (Positive)** or **Empty (Negative)**. Four outcomes are possible:

- **TP** (True Positive): person present and correctly detected.
- **TN** (True Negative): no person present and system correctly silent.
- **FP** (False Positive): no person present but alarm raised (*false alarm*).
- **FN** (False Negative): person present but missed by the system (*missed detection*).

3.2 Derived Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

Accuracy (Eq. 1) is the overall fraction of correctly classified frames. On a heavily imbalanced dataset, accuracy is susceptible to optimistic inflation: a detector that always fires achieves $\approx 98\,\%$ accuracy while missing every empty frame.

Precision (Eq. 2) is the fraction of predicted positives that are genuinely positive. High precision implies few false alarms. Excessive false positives cause *alert fatigue*, leading staff to disable or ignore the system.

Recall (Eq. 3) is the fraction of genuine positives that are detected. High recall implies few missed detections. Recall is the **primary safety metric**: a missed intruder, fallen patient, or fire event is a direct system failure.

F1-Score (Eq. 4) is the harmonic mean of precision and recall. It balances both objectives and is used as the headline metric because of the class imbalance.

3.3 Success Criteria

Two criteria are applied throughout this report:

Criterion A — Frame-level presence (no IoU):
The detector is credited with a TP whenever it outputs *any* bounding box on a person-present frame, regardless of spatial accuracy. This criterion answers: *"Did the system know someone was in the room?"* It is appropriate for simple presence-only alarming (e.g., restricted-area breach detection).

Criterion B — Localisation (IoU > 0.5):
A frame is counted as TP only if at least one predicted box overlaps a ground-truth box by more than $50\,\%$ Intersection-over-Union (IoU). This criterion answers: *"Did the system correctly locate the person?"* Accurate localisation is a prerequisite for passing bounding boxes downstream to contact detection algorithms.

4 Algorithms

Table 2: Summary of evaluated algorithms.

Algorithm	Type	Training	Description
Adaptive Threshold	Rule-based CV	None	Adaptive Gaussian thresholding, morphological closing, and geometric blob filtering (area, solidity, aspect ratio).
HOG + SVM	Classical ML	LinearSVC	Sliding window (14×6 px) with Histogram of Oriented Gradients features; Linear SVM classifier; single scale.
HOG + SVM + Pyramid	Classical ML	Same as above	Identical trained SVM weights; inference repeated at scales $\{0.75\times, 1.00\times, 1.25\times, 1.50\times\}$; unified NMS across all scales.
MobileNet-SSD	Deep learning	PyTorch (CUDA)	Micro-MobileNet backbone with SSD anchor regression heads; 30 training epochs; RTX 2060 Super.

5 Results — Criterion A: Frame-Level Presence

Under Criterion A, a detection anywhere on a person-present frame is counted as correct.

5.1 Confusion Matrices

Adaptive Threshold

	Pred: Human	**Pred: Empty**
Truth: Human (608 frames)	**TP** = 591	**FN** = 17
Truth: Empty (13 frames)	**FP** = 1	**TN** = 12

HOG + SVM (single scale)

	Pred: Human	**Pred: Empty**
Truth: Human (608 frames)	**TP** = 585	**FN** = 23
Truth: Empty (13 frames)	**FP** = 3	**TN** = 10

MobileNet-SSD

	Pred: Human	**Pred: Empty**
Truth: Human (608 frames)	**TP** = 593	**FN** = 15
Truth: Empty (13 frames)	**FP** = 0	**TN** = 13

5.2 Metric Summary

Table 3: Criterion A — test-set metrics (frame-level presence, no IoU requirement).

Algorithm	Acc	Prec	Rec	F1	TP	TN	FP	FN
Adaptive Threshold	97.1%	99.8%	97.2%	**98.5%**	591	12	1	17
HOG + SVM	95.8%	99.5%	96.2%	**97.8%**	585	10	3	23
MobileNet-SSD	97.6%	**100.0%**	**97.5%**	**98.8%**	593	13	0	15

5.3 Interpretation

Under the lenient criterion, **all three algorithms perform well**, with F1-scores between 97.8 % and 98.8 %. The results demonstrate that *detecting the presence* of a person in a thermal frame is a tractable problem even for a parameter-free, rule-based approach.

MobileNet-SSD leads at F1 = 98.8 % and is the only algorithm to produce **zero false positives** — it never raised an alarm on an empty room. Its 15 missed detections are mostly frames with partial occlusion or atypical thermal profiles.

Adaptive Threshold is effectively tied (F1 = 98.5 %) with no training cost. Its single false positive originates from a thermal reflection artefact in the empty-room recording.

HOG + SVM ranks third at F1 = 97.8 %, with 3 false alarms: the sliding window found background thermal gradients that matched its learnt person template.

The conclusion from Criterion A alone would be misleadingly optimistic. The next section exposes the true differences between algorithms.

6 Results — Criterion B: Localisation (IoU > 0.5)

Under Criterion B, a frame only counts as a TP if the predicted bounding box overlaps the ground-truth annotation by IoU > 0.5. Frames where the person was detected but the box was misplaced become false negatives.

6.1 Without Scale Pyramid

Adaptive Threshold

	Pred: Human	Pred: Empty
Truth: Human (608 frames)	**TP** = 423	**FN** = 185
Truth: Empty (13 frames)	**FP** = 1	**TN** = 12

HOG + SVM (single scale)

	Pred: Human	Pred: Empty
Truth: Human (608 frames)	**TP** = 246	**FN** = 362
Truth: Empty (13 frames)	**FP** = 3	**TN** = 10

MobileNet-SSD

	Pred: Human	Pred: Empty
Truth: Human (608 frames)	**TP** = 549	**FN** = 59
Truth: Empty (13 frames)	**FP** = 0	**TN** = 13

6.2 With Scale Pyramid (HOG + SVM)

The scale pyramid runs the same trained SVM weights on four rescaled copies of each frame (0.75×, 1.00×, 1.25×, 1.50×), projects all detections back to original coordinates, and performs a single unified NMS pass.

HOG + SVM + Pyramid

	Pred: Human	Pred: Empty
Truth: Human (608 frames)	**TP** = 376	**FN** = 232
Truth: Empty (13 frames)	**FP** = 12	**TN** = 1

6.3 Metric Summary

Table 4: Criterion B — test-set metrics (IoU > 0.5 localisation requirement).

Algorithm	**Acc**	**Prec**	**Rec**	**F1**	**TP**	**TN**	**FP**	**FN**
Adaptive Threshold	70.0%	99.8%	69.6%	**82.0%**	423	12	1	185
HOG + SVM	41.2%	98.8%	40.5%	**57.4%**	246	10	3	362
HOG + SVM + Pyramid	60.7%	96.9%	61.8%	**75.5%**	376	1	12	232
MobileNet-SSD	90.5%	**100.0%**	**90.3%**	**94.9%**	549	13	0	59

6.4 Interpretation

The localisation criterion separates the algorithms dramatically: nearly all new false negatives are frames where the system *did* detect a person but placed the box inaccurately.

MobileNet-SSD remains the clear leader at F1 = 94.9 % (−3.9 pp from Criterion A). Its SSD anchor regression heads are explicitly trained to minimise bounding box error, making it the only algorithm that genuinely *localises* persons rather than merely detecting their presence. The 59 false negatives are concentrated in scenes with atypical postures (falling, crouching) and dense multi-person occlusion. Crucially, **zero false alarms** are produced.

Adaptive Threshold drops to F1 = 82.0 % (−16.5 pp). The morphological closing reliably finds the hottest blob in a frame, but the resulting contour bounding rectangle is seldom tight enough to clear IoU = 0.5 against a manually drawn annotation. The 185 false negatives are almost entirely frames where a person *was* detected but the blob shape diverged from the GT box.

HOG + SVM (single scale) collapses to F1 = 57.4 % (−40.4 pp) — the worst localisation result. The root cause is the **fixed sliding window** (14×6 px on a 32×24 sensor). When a person appears at a scale or aspect ratio different from the training window, the best achievable IoU is bounded well below 0.5 regardless of SVM confidence: detection occurs, but localisation does not.

HOG + SVM + Pyramid partially recovers to F1 = 75.5 % (+18.1 pp over single scale). Running inference at four scales allows the fixed window to match persons across a range of apparent distances, converting 130 additional false negatives to true positives. However, the pyramid also introduces a significant false-alarm regression: FP rises from 3 to 12, and true negatives on empty frames fall from 10 to 1. The multi-scale sweep provides more recall opportunities on positive frames at the cost of substantially reduced selectivity on empty backgrounds.

7 Consolidated Comparison

Table 5 presents all results side-by-side for direct comparison across both criteria and all algorithms.

Table 5: Complete metric comparison — test set, all algorithms, both criteria.

Algorithm	**Criterion**	**Accuracy**	**Precision**	**Recall**	**F1**
Adaptive Threshold	No IoU	97.1%	99.8%	97.2%	98.5%
	IoU > 0.5	70.0%	99.8%	69.6%	82.0%
HOG + SVM	No IoU	95.8%	99.5%	96.2%	97.8%
	IoU > 0.5	41.2%	98.8%	40.5%	57.4%
HOG + SVM + Pyramid	IoU > 0.5	60.7%	96.9%	61.8%	75.5%
MobileNet-SSD	No IoU	97.6%	100.0%	97.5%	98.8%
	IoU > 0.5	90.5%	100.0%	90.3%	**94.9%**

The F1 drop from Criterion A to Criterion B quantifies each algorithm's *localisation gap*:

Table 6: Localisation gap: F1 drop from Criterion A to Criterion B.

Algorithm	**F1 (no IoU)**	**F1 (IoU > 0.5)**	**Drop**
MobileNet-SSD	98.8%	94.9%	−3.9 pp
Adaptive Threshold	98.5%	82.0%	−16.5 pp
HOG + SVM	97.8%	57.4%	−40.4 pp
HOG + SVM + Pyramid	—	75.5%	−22.3 pp vs. no-IoU

8 Conclusions

8.1 Presence Detection (Criterion A)

All three base algorithms achieve F1 $\geq$ 97.8 % under the frame-level presence criterion. For applications requiring only a binary room-occupancy signal (e.g., restricted-area breach detection), even the parameter-free Adaptive Threshold detector is sufficient, with the advantage of requiring no training data or GPU resources.

8.2 Localisation (Criterion B)

Only **MobileNet-SSD** meets a practical localisation threshold, achieving F1 = 94.9 % with zero false alarms. It is the recommended algorithm for any downstream task that consumes bounding boxes — most importantly, contact detection, where the foot-point projection from each person's bounding box is the input to the multi-view fusion stage.

Adaptive Threshold achieves F1 = 82.0 % and may be acceptable as a lightweight fallback, but its 30 % false-negative rate on localisation means that roughly one in three person-present frames will not produce a usable bounding box.

HOG + SVM, even with scale pyramid (F1 = 75.5 %), remains 19 pp below MobileNet-SSD. The pyramid improves recall by +18 pp but introduces unacceptable false-alarm behaviour on empty frames (92 % false-alarm rate). Further improvement would require **hard negative mining** to suppress background misfires, followed by a **bounding-box regression layer** to correct residual window-position errors.

8.3 Recommendation

- **Primary detector:** MobileNet-SSD — best F1, zero false positives, suitable for bounding-box–dependent downstream tasks.
- **Lightweight fallback:** Adaptive Threshold — no training required, adequate for presence-only alarming.
- **HOG + SVM:** Not recommended for localisation tasks in its current form; viable only as a presence-only detector without the pyramid.